

TryNex Lifestyle

Bangladesh's #1 Custom Apparel & Lifestyle Brand

Full Technical Documentation

A to Z Complete Tech Stack, Architecture & Feature Reference

Version 2.0 • May 2026 • Production-Ready

React 19 + Vite 7 + Express 5 + PostgreSQL + Drizzle ORM + Three.js

Cloudflare R2 • Upstash Redis • Replit Platform • pnpm Monorepo

TypeScript

React 19

Vite 7

Express 5

PostgreSQL

Three.js

Tailwind v4

Framer Motion

Drizzle ORM

Upstash Redis

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01 — Project Overview & Business Context

What is TryNex Lifestyle?

TryNex Lifestyle is a full-stack e-commerce platform built for a Bangladesh-based custom apparel and lifestyle brand. Customers can design, customise, and order personalised T-shirts, hoodies, mugs, caps, water bottles, and gift hampers — delivered nationwide across all 64 districts with Cash on Delivery support.

Business Highlights

- 5,000+ happy customers served across Bangladesh
- 24-hour production turnaround on standard orders
- 320 GSM premium fabric quality
- Cash on Delivery (COD) available on all orders
- Nationwide delivery — all 64 districts
- Online payments via bKash, Nagad, uPay
- 3D Design Studio — real-time product personalisation
- Gift Hampers — curated and custom builds
- Referral system with discount rewards
- Admin panel with full order lifecycle management

Core Platform Capabilities

- Buyer storefront: listing, detail pages, cart, checkout, order tracking
- Design Studio: upload/text/templates on 3D models (T-shirts, hoodies, mugs, caps, bottles)
- Admin dashboard: order management, product CRUD, blog, SEO, promo codes, analytics
- Customer accounts: registration, login, order history, messaging, notification bell
- Live order chat between customer and admin
- Notification bell — real-time order status alerts (NEW)
- Blog / CMS for SEO content
- Gift hamper builder (curated + custom)
- Social proof: testimonials, star reviews

02 — Complete Tech Stack Reference

Frontend Stack

Layer	Technology	Version	Purpose
Framework	React	19.1.0	UI component tree, concurrent rendering
Build Tool	Vite	7.3.3	HMR dev server, ESM bundling
Language	TypeScript	5.8.3	Type-safe code across all packages
Styling	Tailwind CSS	v4 (oxide)	Utility-first CSS, zero-runtime
Routing	wouter	3.7.1	Lightweight client-side routing
Animation	Framer Motion	12.x	Spring physics, gestures
3D Engine	Three.js	r176	WebGL 3D rendering
3D React	React Three Fiber	9.x	Declarative Three.js in React
3D Helpers	Drei	10.x	Camera controls, environment
Forms	React Hook Form	7.x	Uncontrolled form handling
Validation	Zod	3.x	Schema validation
Data Fetch	TanStack Query	5.x	Server state, caching, refetch
Icons	Lucide React	0.511.x	SVG icon library

Backend Stack

Layer	Technology	Version	Purpose
Runtime	Node.js	24.x	V8 engine, ES2024, async/await
Framework	Express	5.x	HTTP routing, middleware pipeline
Language	TypeScript	5.8.3	Compiled to ESM via esbuild
ORM	Drizzle ORM	0.40.x	Type-safe SQL, schema-first migrations
Validation	Zod	3.x	Request body validation
Logging	Pino	9.x	Structured JSON logging
Auth JWT	jsonwebtoken	9.x	Customer + Admin JWT signing
Scheduler	In-process cron	custom	Daily alerts, keep-alive pings
Image AI	Transformers.js	3.x	In-browser ML background removal

Infrastructure & Services

Service	Provider	Purpose	Fallback
Database	Replit PostgreSQL	Primary relational store	Neon DB failover chain (3 URLs)
Cache	Upstash Redis REST	Distributed response cache	In-process Map
Object Storage	Cloudflare R2	Images, mockups, assets	Local disk (dev)
CDN	Cloudflare	R2 public CDN edge	Direct R2 URL
Hosting	Replit	Managed container + proxy	Render.com (alt)

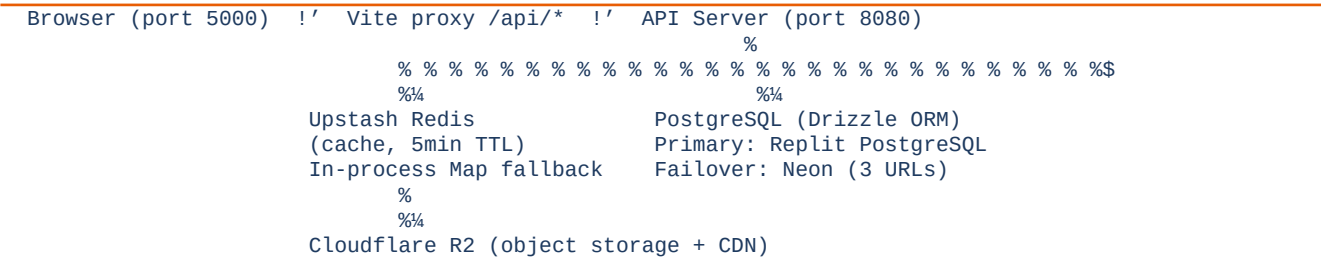
Service	Provider	Purpose	Fallback
CI/CD	GitHub Actions	Auto-deploy on push to main	Manual trigger
Version Control	GitHub	Source of truth	Local git

03 — Monorepo Architecture

pnpm workspace structure

```
workspace-root/
% % % artifacts/
%   % % % trynex-storefront/      # React 19 + Vite 7 (port 5000)
%   %   % % % src/
%   %   %   % % % pages/          # Route-level components (20+ pages)
%   %   %   % % % components/     # Shared UI (Navbar, Footer, CartDrawer...)
%   %   %   % % % context/        # CartContext, AuthContext, SiteSettingsContext
%   %   %   % % % hooks/          # Custom React hooks
%   %   %   % % % lib/            # Utils, API client, helpers
%   % % % api-server/             # Express 5 + Node 24 (port 8080)
%   %   % % % src/
%   %   %   % % % routes/         # 30+ Express route files
%   %   %   % % % lib/           # Auth, logger, storage, scheduler
%   % % % api-worker/            # Cloudflare Worker (Hono) – alt deploy
%   % % % mockup-sandbox/        # Isolated UI dev/preview (port 8081)
% % % lib/
%   % % % db/                    # Drizzle schema + 17 migrations (0001-0016)
%   % % % api-spec/              # OpenAPI 3.1 spec + generated TS types
%   % % % api-zod/               # Generated Zod validators
%   % % % api-client-react/      # TanStack Query hooks
% % % scripts/                  # Seed data, CI helpers
% % % proxy-8084.mjs            # Replit preview proxy
```

Request Data Flow



04 — Frontend — React + Vite Storefront

All Pages & Routes

Route	Component	Description
/	Home.tsx	Hero, products, testimonials, blog previews
/shop	Shop.tsx	Product grid, filters, search, pagination
/products/:slug	ProductDetail.tsx	Colour/size picker, inline validation, 3D preview
/design-studio	DesignStudio.tsx	3D Design Studio (4,500+ lines)
/cart	Cart.tsx	Full cart with 3D preview toggle, re-edit
/checkout	Checkout.tsx	3-step: Delivery !' Payment !' Review
/track-order	TrackOrder.tsx	10-step timeline, courier info, live chat
/account	Account.tsx	Orders, messages, profile, notifications
/hampers	Hampers.tsx	Gift hamper builder (curated + custom)
/blog/:slug	BlogPost.tsx	SEO-optimised blog content
/referral	Referral.tsx	Referral dashboard + sharing
/admin/*	Admin panel	Full admin dashboard (auth-gated)

Key React Contexts

- CartContext — cart items, quantities, totals; persisted to localStorage
- AuthContext — customer JWT, login/logout, token refresh
- SiteSettingsContext — global settings from /api/settings (logo, contact, colors)

Performance Techniques

- React.lazy() + Suspense route-level code splitting (Three.js isolated to Design Studio route)
- TanStack Query stale-while-revalidate with 5min cache
- Tailwind CSS v4 Oxide engine — zero dead CSS in production builds
- Image lazy load + WebP with responsive srcset
- Vite manual chunk splitting: vendor, framer-motion, three.js in separate bundles

05 — Backend — Express API Server

Server Bootstrap Sequence

- Express 5 with native async error propagation
- Helmet.js: CSP, HSTS, X-Frame-Options, X-Content-Type-Options
- CORS: explicit allowlist for Replit preview + production domains
- Rate limiting: 200 req/15min global, 20/15min on auth endpoints
- Pino HTTP logger with request/response timing
- runMigrations() at startup — applies all pending Drizzle SQL migrations
- In-process scheduler: daily (9am BST), low-stock (10am & 8pm), keep-alive (14min)

Route Modules (30 files)

File	Method	Purpose
auth.ts	POST /api/auth/*	Customer register, login, refresh, profile
orders.ts	GET/POST /api/orders/*	Place order, track, admin CRUD, status + notifications
products.ts	GET/POST /api/products/*	Product catalog CRUD, search, filters
notifications.ts	GET/PATCH /api/notifications/*	Customer notification bell (NEW)
orderMessages.ts	GET/POST /api/orders/:id/messages	Customer-admin live chat
reviews.ts	GET/POST /api/reviews/*	Product reviews, ratings, moderation
promoCode.ts	POST /api/promo/*	Promo code validation, usage tracking
referrals.ts	GET/POST /api/referrals/*	Referral links and discount application
hampers.ts	GET/POST /api/hampers/*	Gift hamper packages and custom builder
storage.ts	POST /api/storage/*	Cloudflare R2 presigned upload URLs
admin.ts	GET/* /api/admin/*	Dashboard stats, bulk actions
blog.ts	GET/POST /api/blog/*	Blog post CMS with SEO fields
settings.ts	GET/PUT /api/settings/*	Site-wide config key-value store
ai.ts	POST /api/ai/*	AI image background removal, upscale
seo.ts	GET /sitemap.xml, /robots.txt	SEO utilities

06 — Database Schema & Migrations

Connection Strategy — Multi-DB Failover

DATABASE_URL	!' Replit PostgreSQL (primary, auto-provisioned)
DATABASE_URL_MAIN	!' Neon DB failover #1
DATABASE_URL_TRYNEX_DB	!' Neon DB failover #2
DATABASE_FAILOVER	!' Neon DB failover #3

All 17 Tables

Table	Key Columns	Purpose
users	id, email, phone, passwordHash, createdAt	Customer accounts
orders	id, orderNumber, customerId, status, total, items(jsonb)	All orders
order_items	orderId, productId, qty, price, size, color, customNote	Line items
products	id, slug, name, price, discountPrice, colors, sizes, images	Product catalog
categories	id, slug, name, parentId, position	Category tree
notifications	id, customerId, type, title, message, orderId, read, createdAt	NEW — order alerts
order_messages	id, orderId, fromAdmin, message, createdAt	Customer-admin chat
reviews	id, productId, userId, rating, body, approved	Product reviews
promo_codes	id, code, discount, type, maxUses, usedCount, expiresAt	Promotions
referrals	id, code, referrerId, rewardDiscount, usedCount	Referral program
hamper_packages	id, name, basePrice, items(jsonb), active	Gift hamper catalog
blog_posts	id, slug, title, content, seoTitle, seoDesc, publishedAt	Blog CMS
settings	key, value	Site-wide config key-value store
testimonials	id, name, message, rating, approved	Social proof
admin_activity_logs	id, adminId, action, entityType, entityId, meta	Audit trail
abandoned_carts	id, sessionId, items(jsonb), email, recoveredAt	Cart recovery
drafts	id, userId, content(jsonb), type, createdAt	Studio drafts

Migrations Applied (0001–0016)

- 0001 — initial_schema: users, products, orders, categories
- 0002 — add_blog: blog_posts table
- 0003 — add_reviews: product reviews
- 0004 — settings_table: key-value config
- 0005 — add_promos: promo codes
- 0006 — add_referrals: referral program
- 0007 — add_hampers: gift hamper packages
- 0008 — add_testimonials: customer testimonials

- 0009 — order_messages: customer-admin messaging
- 0010 — activity_logs: admin audit trail
- 0011 — abandoned_carts: cart recovery
- 0012 — drafts: Design Studio draft saving
- 0013 — color_variants: product colour variant metadata
- 0014 — admin_activity: extended activity log fields
- 0015 — safe_product_columns: nullable column additions
- 0016 — notifications: customer notification bell (NEW in this release)

07 — Full API Endpoints Reference

Authentication

Method	Endpoint	Auth	Description
POST	/api/auth/register	Public	Create customer account
POST	/api/auth/login	Public	Login, returns JWT access + refresh tokens
POST	/api/auth/refresh	JWT	Refresh expired access token
GET	/api/auth/me	JWT	Get current customer profile
PUT	/api/auth/profile	JWT	Update name/phone/address

Orders

Method	Endpoint	Auth	Description
POST	/api/orders	Public	Place new order (catalog + studio + hamper items)
POST	/api/orders/track	Public	Track by order number + email/phone
GET	/api/orders/my	JWT	Customer order history with pagination
PUT	/api/orders/:id/payment-info	JWT	Submit payment reference info
GET	/api/orders/my/messages/unread-count	JWT	Unread chat message count
GET	/api/admin/orders	Admin	All orders with status/date filters
PATCH	/api/admin/orders/:id/status	Admin	Update status, triggers notification

Notifications (NEW)

Method	Endpoint	Auth	Description
GET	/api/notifications	JWT	Paginated notification list (page, limit)
GET	/api/notifications/unread-count	JWT	Unread count for bell badge
PATCH	/api/notifications/:id/read	JWT	Mark one notification as read
PATCH	/api/notifications/mark-all-read	JWT	Mark all notifications as read

Products, Reviews, Promo

Method	Endpoint	Auth	Description
GET	/api/products	Public	All products, category/search/sort filters
GET	/api/products/:id	Public	Single product with colour variants
GET	/api/categories	Public	Full category tree
GET	/api/reviews/:productId	Public	Approved reviews for a product
POST	/api/reviews	JWT	Submit a product review
POST	/api/promo/validate	JWT	Validate and apply promo code

Design Studio & Storage

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Method	Endpoint	Auth	Description
POST	/api/storage/upload	JWT	Upload image to Cloudflare R2, returns CDN URL
POST	/api/ai/remove-bg	JWT	AI background removal (Transformers.js in-browser)
POST	/api/ai/upscale	JWT	AI image upscaling 2x/4x
GET	/api/drafts	JWT	Load saved Design Studio drafts
POST	/api/drafts	JWT	Save Design Studio draft (JSON layers state)

08 — 3D Design Studio

Overview

The Design Studio (DesignStudio.tsx, 4,500+ lines) is the centrepiece of TryNex. It provides real-time in-browser product customisation with a full 3D preview using React Three Fiber. Customers upload artwork, add text, apply templates, and see their design live on a rotating 3D model before ordering.

Supported Products & Zones

Product	Design Zones	Colour Options	3D Model
T-Shirt	Front + Back	8 variants	tshirt.glb
Hoodie	Front + Back + Sleeves	6 variants	hoodie.glb
Long-Sleeve Shirt	Front + Back	5 variants	longsleeve.glb
Mug	Left Side / Right Side / Full Wrapping	3 variants	mug.glb
Cap	Front embroidery zone	4 variants	cap.glb
Water Bottle	Wrap zone (carabiner)	2 variants	bottle.glb

Features

- Image upload — drag & drop, file picker, paste from clipboard
- Text layers — 20+ font families, size, weight, colour, shadow, stroke
- AI Background Removal — Transformers.js, runs entirely in-browser (no server cost)
- AI Upscale — 2× / 4× image upscaling
- Templates — pre-designed layouts for quick start
- Layers panel — show/hide, lock, reorder, duplicate (NEW), delete
- Main floating toolbar (NEW) — Upload, Text, Templates, Layers, Undo, Redo, Clear buttons
- Full undo/redo — Ctrl+Z / Ctrl+Y keyboard shortcuts
- Multi-face — animated switch between Front / Back / Sleeves
- Print zone L-bracket corner guides showing bleed area
- Zoom/pan — pinch-to-

zoom mobile, scroll wheel
desktop

- Snap guides —
horizontal and vertical
centre-snap lines
- Draft auto-save to
localStorage + explicit
Save Draft button
- Export to cart —
canvas thumbnail
generated, added as
studio item
- 3D product viewer —
rotate/zoom live 3D model
(React Three Fiber)

3D Rendering Pipeline

```
GLB models  !'  useGLTF (Drei, draco compression)
!'"
Canvas design  !'  exportToBlob()  !'  THREE.Texture
!'"
3D Mesh TextureMap  +  HDR environment lighting
!'"
OrbitControls (touch-friendly rotate/zoom)
!'"
Rendered in <Canvas> (React Three Fiber)
```

09 — Checkout & Payment Flow

3-Step Checkout (Redesigned)

Step	Content	Validation
Step 1: Delivery	Full name, phone, email, address (district, thana area street) notes	All required; phone format; email format
Step 2: Payment	Method (bKash/Nagad/uPay/COD), mode (Full/15% Advance) transaction ID	Payment ref required for online; COD skips
Step 3: Review	Order summary, items, totals, promo code, Place Order CTA	Promo validated server-side before submit

Payment Methods

- Cash on Delivery (COD) — pay on receipt, no upfront deposit required
- bKash — Bangladesh largest mobile financial service
- Nagad — government-backed digital wallet
- uPay — digital wallet
- Payment modes: Full payment OR 15% advance deposit
- Trust badges shown below Place Order button

Post-Order Server Logic

- All prices recomputed server-side from DB — client totals never trusted
- Studio item prices: server-side lookup from settings table
- Promo discounts: server-side validation, client discount ignored
- Order placed !' status: pending !' admin Telegram notification
- Admin updates status !' notifications table entry created for customer
- Shipped: courier name + tracking URL stored, visible on Track Order page

10 — Notifications System (New in This Release)

Backend Architecture

- Table: notifications (id, customerId, type, title, message, orderId, read, createdAt)
- Migration 0016_notifications.sql applied automatically at server start
- Types: order_update, message, general
- Triggers: admin updates order status !' notification created; admin sends message !' notification created
- All endpoints protected by customer JWT auth (requireCustomer middleware)

API Endpoints

Method	Endpoint	Description
GET	/api/notifications?page=1&limit=20	Paginated notification list for logged-in customer
GET	/api/notifications/unread-count	Returns { count: N } for bell badge
PATCH	/api/notifications/:id/read	Mark one notification as read
PATCH	/api/notifications/mark-all-read	Mark all unread notifications read

Frontend Bell (Navbar.tsx)

- Bell icon visible only when customer is logged in
- Red pulsing badge shows live unread count
- Dropdown on click: list of recent notifications with type-colour icons
- Blue = order update, Green = message, Orange = general
- Auto-marks-all-read when dropdown is opened
- Polls /api/notifications/unread-count every 30 seconds
- Click-outside closes dropdown;
- "Mark all read" button in header
- "View Details" links navigate to relevant account section

11 — Order Tracking System

10-Step Status Timeline

Status Key	Display Label	Description
pending	Order Placed	Order received, awaiting admin confirmation
confirmed	Confirmed	Admin has confirmed the order
design_review	Design Review	Custom design being reviewed by team
design_approved	Design Approved	Design approved, production starting
in_production	In Production	Being manufactured / printed
quality_check	Quality Check	Quality inspection underway
ready_to_ship	Ready to Ship	Packed and awaiting courier pickup
shipped	Shipped	Sent to delivery partner
out_for_delivery	Out for Delivery	With last-mile delivery agent
delivered	Delivered	Successfully delivered to customer

Courier Tracking & Timeline

- Courier name + tracking number displayed when admin sets on order
- External tracking link opens in new tab if tracking URL provided
- Status timeline events reconstructed from admin_activity_logs table
 - Each event shows: icon, status label, timestamp (date + time)
 - Customer messaging (live chat) integrated on same Track Order page

12 — Cart System

CartContext (State Management)

- Persisted to localStorage — survives page refresh and tab close
- Cart items carry: id, productId, name, price, qty, size, color, imageUrl, customNote, customImages, isStudio
- Computed values: subtotal, total, item count (memoised)
- Actions: addItem, removeItem, changeQuantity, clearCart
- Studio items carry design thumbnail (data URL converted to R2 path at order-time)

Cart Drawer (Slide-Over)

- "Re-edit" button for Design Studio items — reopens studio with saved session
- Shows product thumbnail / design preview for all items
- Size + Color labels for clarity
- Order total with shipping calculation
- Framer Motion spring slide-in animation from right side

Cart Page (/cart)

- "View in 3D" toggle for studio items — CartViewer3D component embedded
- Empty state with Browse Collection CTA + recently viewed products
- Promo code input with real-time server validation
- Quantity steppers with instant total update (no debounce or race conditions)

13 — Admin Panel

Access & Auth

Visit /admin/login. Authenticate with ADMIN_PASSWORD. Admin JWT is separate from customer JWT (ADMIN_JWT_SECRET). Token stored in sessionStorage — auto-clears on tab close.

Admin Sections

Section	Key Actions
Dashboard	Live stats: orders, revenue, pending count, low-stock alerts
Orders	Filter by status/date/search; bulk update; export CSV
Order Detail	Items, customer info, messages, status update, courier tracking input
Products	Add/edit/delete; image upload to R2; colour variants; stock
Categories	Category tree management with parent/child relationships
Blog	Post editor with rich text, SEO fields, auto-slug generation
Promo Codes	Create/deactivate codes; track usage count and limits
Reviews	Approve/reject product reviews; star rating moderation
Hampers	Manage curated hamper packages and pricing
Testimonials	Approve/reject customer testimonials
Settings	Site name, logo, WhatsApp, contact, shipping thresholds
SEO	Meta tags, OpenGraph, structured data overrides per page
Activity Log	Full audit trail of all admin actions with timestamps
Deployment	Trigger manual deploys, server health check

14 — Storage (Cloudflare R2) & Cache (Upstash Redis)

Cloudflare R2 Storage

- S3-compatible API — AWS SDK v3 client-side presigned uploads
- Single bucket: trynex-assets with path prefixes
- Path structure: /products/, /studio/, /blog/, /avatars/, /mockups/
- Public CDN access via Cloudflare edge network (global PoPs)
- Presigned upload URLs: 15-minute expiry, content-type locked to image/*
- Studio: data-URL thumbnails converted to R2 paths at order-creation time
- Env vars: R2_ACCESS_KEY_ID, R2_SECRET_ACCESS_KEY, R2_ENDPOINT, R2_BUCKET

Upstash Redis Cache

- HTTP REST API — works in serverless/edge without raw TCP connections
- Default TTL: 5 minutes for product listings and settings
- Cache keys: products:all, settings, categories, public-stats
- In-process Map fallback: if Redis unavailable, memory cache used for server lifetime
- Cache invalidated on admin CRUD mutations (products/settings/categories)
- Env vars: UPSTASH_REDIS_REST_URL, UPSTASH_REDIS_REST_TOKEN

15 — Authentication

Customer Auth (JWT)

- Access token (15 min) + Refresh token (30 days) pair
- Tokens stored in localStorage on the storefront
- Passwords hashed: bcrypt 12 rounds (never stored plain)
- extractCustomerToken() reads Bearer header or cookie
- verifyCustomerToken() decodes and validates signature
- Env: JWT_SECRET (minimum 32 characters, rotatable)

Admin Auth (Separate JWT)

- Signed with ADMIN_JWT_SECRET — completely separate from customer keys
- Login: username + ADMIN_PASSWORD secret
- All /api/admin/* routes protected by requireAdmin middleware
- Token stored in sessionStorage (auto-clears on tab close for security)
- Env: ADMIN_JWT_SECRET, ADMIN_PASSWORD, ADMIN_SECRET_PASSWORD

Guest Order Support

- Customers can order without an account (guest checkout)
- Guest session: phone + email combination identifies the customer
- Order tracking: works via order number + phone/email (no login required)

16 — Environment Variables & Secrets

All secrets managed via Replit Secrets (never in .env files or source code). Set in the Replit Secrets tab. Never commit secrets to git.

Variable	Required	Description
DATABASE_URL	Yes	Replit PostgreSQL primary DB (auto-provisioned)
DATABASE_URL_MAIN	Yes	Neon DB failover #1
DATABASE_URL_TRYNEX_DB	No	Neon DB failover #2
DATABASE_FAILOVER	No	Neon DB failover #3
JWT_SECRET	Yes	Customer JWT signing key (min 32 chars)
ADMIN_JWT_SECRET	Yes	Admin JWT signing key (separate from customer)
ADMIN_PASSWORD	Yes	Admin panel login password
ADMIN_SECRET_PASSWORD	No	Secondary admin access password
R2_ACCESS_KEY_ID	Yes	Cloudflare R2 access key ID
R2_SECRET_ACCESS_KEY	Yes	Cloudflare R2 secret access key
R2_ENDPOINT	Yes	R2 S3-compatible endpoint URL
R2_BUCKET	Yes	R2 bucket name (e.g. trynex-assets)
UPSTASH_REDIS_REST_URL	No	Upstash Redis URL (in-mem fallback if absent)
UPSTASH_REDIS_REST_TOKEN	No	Upstash Redis auth token
GITHUB_TOKEN	No	GitHub API token for CI/CD operations
GITHUB_PERSONAL_ACCESS_TOKEN	No	GitHub PAT for manual repo push
RENDER_API_KEY	No	Render.com deployment API key
CLOUDFLARE_API_TOKEN	No	Cloudflare API token for Worker deploy
TELEGRAM_BOT_TOKEN	No	Telegram bot for admin order alerts

17 — CI/CD & Deployment

GitHub Actions ([github/workflows/deploy.yml](#))

- Trigger: push to main branch on [github.com/georgelsmith333-hub/trynex-liestyle](#)
- Steps: checkout !' pnpm install !' typecheck (non-blocking || true) !' esbuild !' deploy
- Deployment targets: Cloudflare Worker (api-worker) + Render.com (api-server)
- Both deploy steps use continue-on-error: true (partial deploys allowed)
- CF deploy guarded by CLOUDFLARE_API_TOKEN secret presence check
- Render deploy triggers service deploy via Render API with RENDER_API_KEY

Replit Start Command

```
node proxy-8084.mjs &  
DATABASE_URL="$DATABASE_URL_MAIN" NODE_ENV=development \  
  pnpm --filter @workspace/api-server run dev &  
PORT=5000 API_PORT=8080 NODE_ENV=development \  
  pnpm --filter @workspace/trynex-storefront run dev
```

Deployment Notes

- API server: TypeScript compiled with esbuild !' dist/index.mjs !' Node 24
- Storefront: Vite dev server (development) / vite build + preview (production)
- Vite proxies all /api/* to :8080 — no CORS issues in development
- Replit proxy-8084 routes public traffic to port 5000 via mTLS
- Suggest Deploy creates .replit.app subdomain with TLS + health checks

18 — Performance, SEO & Security

Frontend Performance

- Three.js loaded only on Design Studio route (lazy chunk — ~2MB isolated)
- React.lazy() splits admin panel out of main bundle
- Vite chunk splitting: vendor, framer-motion, three.js in separate bundles
- TanStack Query: stale-while-revalidate, 5min cache, background refetch on window focus
- Tailwind CSS v4 Oxide engine — zero unused CSS in production
- Images: WebP with fallback, lazy load, responsive srcset
- Fonts: Google Fonts preconnect + font-display: swap (zero FOIT)

SEO

- SEOHead.tsx: per-page title, meta description, canonical URL
- OpenGraph + Twitter Card meta for social sharing previews
- JSON-LD structured data: Product, BreadcrumbList, Organization
- Dynamic sitemap.xml — /api/sitemap — includes all products + blog posts
- robots.txt with admin panel exclusion
- Blog: rich SEO content, keyword-optimised slugs, publishedAt meta

Security Architecture

- Helmet.js: CSP, HSTS, X-Frame-Options, X-Content-Type-Options headers
- Rate limiting: 200 req/15min global, 20/15min on auth endpoints
- CORS: explicit origin allowlist only
- Zod schema validation on all mutation endpoints
- Drizzle ORM parameterised queries — no SQL injection surface
- React escapes all rendered content — no dangerouslySetInnerHTML usage
- JWT secrets: minimum 32 chars, admin/customer keys are separate
- Presigned R2 upload URLs: 15min expiry, content-type locked
- All prices recomputed server-side — client totals ignored
- bcrypt 12 rounds for all stored passwords

TryNex Lifestyle

Full Technical Documentation — End of Document

Version 2.0 • May 2026 • Confidential & Proprietary

Repository: github.com/georgelsmith333-hub/trynex-liestyle

Platform: Replit | Stack: React 19 + Express 5 + PostgreSQL

Storage: Cloudflare R2 | Cache: Upstash Redis

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